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Pre-Frac Treatment Pressure Analysis in the Huntersville Chert and Oriskany Sandstone

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Abstract

Pre-frac pressure analyses were done on the Oriskany or Huntersville Chert/ Oriskany combination in wells located in Ohio and Pennsylvania. The pre-treatment pump-in tests were designed to pump several thousand gallons of chemically treated fresh water at multiple pump rates. Upon pump shutdown at the end of this “mini-frac”, the wellhead pressure decline was monitored for an extended time period. Using a commercially available computer software program, the pump rate vs. pressure data and the pressure decline data were analyzed. Pressure decline analysis incorporated a Horner plot for reservoir pressure and permeability along with regression analyses including the Nolte G Function to identify normal leak-off or pressure dependent leak-off and fracture height recession.

Subsequent to the above analyses, hydraulic fracturing treatments were performed on the wells. The treatment designs were tailored to each individual well incorporating the information obtained from the mini fracs about reservoir properties and hydraulic fracture behavior. Use of a Net Pressure Plot of the fracturing treatment was used to validate leakoff behavior identified in the mini fracs.

Introduction

The Oriskany has been a major source of natural gas production in the Appalachian Basin for more than sixty years. In 1949, in sixty-six fields throughout the Appalachian Basin, estimated total reserves for the Oriskany were 1.6 TCF. Subsequent discoveries have increased this figure substantially. The Elk-Poca Field in West Virginia which is the largest Oriskany Field in the basin has produced in excess of 1.0 TCF¹

The overlaying Huntersville Chert is not always present with the Oriskany across the Appalachian Basin and in fact, there is

a coincidence between no Oriskany production and overlaying Chert¹. Diecchio¹ postulates that the highly fractured Chert is a poor cap rock allowing the gas to escape. As a result in some places, the Chert is a better gas reservoir than the Oriskany. The following geological discussion will provide a brief description of the Huntersville Chert and a more detailed discussion of the Oriskany which is the more prolific gas producer across the Appalachian Basin.

The Huntersville Chert is a dense, impure microcrystalline chert interbedded with silicified shale or mudrock. Due to its brittle nature, the Chert was fractured during deformation thus natural fractures are prevalent throughout the Chert. The Chert occurs primarily in the central part of the Appalachian Basin from McKean County, Pennsylvania south to Smyth County, Virginia. The thickest section of Chert (+/- 260') is found in north central West Virginia.

The lower Devonian Oriskany Sandstone as described by Bruner² is an expansive sheet deposit in the Appalachian Basin extending from New York State to Kentucky (Fig. 1). The thickest section of the reservoir is 250 feet to 300+ feet. This section (+/- 10% of the aerial extent of the Oriskany) lies along the Allegheny Front in a six county area. The counties are Somerset and Bedford counties of Pennsylvania, Garrett and Allegany counties of Maryland, and Mineral and Hampshire counties of West Virginia. Nearly 40% of the Oriskany is less than 50 feet thick. Bruner² describes the lithology of the Oriskany as a hybrid sandstone with variable quantities of detrital quartz and sand-sized carbonate detritus. Within the Oriskany, four basic rock units can be characterized.

Beginning at the base of the Oriskany, a brief description of these units or facies is as follows. Facies 1 can be described as basically a limestone section interbedded with fine-grain sandstone. The second facies is a medium-grained clean sandstone with decreasing carbonate matrix. The most widespread and heterogeneous is facies 3. In facies 3 the rock is a mix of bioturbated calcareous sandstones and fossiliferous sandy limestones with argillaceous and organic laminae. Facies 4 is at the top of the Oriskany. Here there is coarse-grained to pebble sized quartz overlain by laminated fine-grain sandstones with grain size increasing vertically.

The Oriskany in general is a fossiliferous quartz arenite cemented with carbonate and silica minerals. Porosity or the lack of porosity is the result of the lithology and cementation combinations in the Oriskany. Bruner² chose four

cores from West Virginia and one from Kentucky as being representative of the porosity and pore types for the Oriskany. Pore types identified were intergranular, solution, recrystallization, intercrystalline, and combination.

Interstitial porosity and pore type diversity increase from east to west in the Oriskany. Correspondingly, much of the gas production has been from western fields. Core analysis in the eastern extent of the Oriskany has shown that for the most part, the interstitial pores have been filled by quartz and iron calcite cement. The high quartz content resulted in hard and brittle rock prone to fracturing during deformation. Gas production in this area is from anticlinal entrapment and fracture porosity². The rock response to hydraulic fracturing and resulting created geometry are also influenced by the natural fracture network in the Oriskany.

Background

In recent history, nearly every Oriskany producer in the Appalachian basin has required a stimulation treatment incorporating a water-based fluid and a propping agent. In order to maximize the likelihood of a successful stimulation treatment, pre-frac diagnostic tests are recommended as a method to help achieve success. Past publications³ of pre-frac diagnostics in the Appalachian Basin have focused on shallower formations.

A search of the literature revealed much has been written about pre-frac diagnostic testing. In 1979 Nolte⁴ proposed fracture pressure decline analysis could provide key information necessary to the design of a successful hydraulic fracturing treatment. In his paper, Nolte defined the G function to represent dimensionless pressure difference. The Oriskany presents a case study of what type problems can be identified by pressure decline analysis incorporating the G function. The G function methodology exhibits unique characteristics for deciding if non-ideal fracture behavior problems result from fluid leakoff out of the created hydraulic fracture into the rock matrix or natural fractures⁵. In low permeability hard sandstone such as the Oriskany, four leakoff types can be identified⁶.

A comparatively small volume minifrac is pumped before the main frac treatment. This minifrac is used to identify these leakoff types. Analysis of the minifrac is done by monitoring pressure drop with time after pump shut down. A plot is created displaying several derivatives vs. Nolte G time. The derivatives of the G function dP/dG and GdP/dG can be used to identify the following:⁵

1. During shut-in, normal leakoff occurs through the formation matrix when the created fracture area is constant. The derivative dP/dG trace will be constant and the GdP/dG will lie on a straight line extrapolated through the origin (Fig. 13).
2. Leakoff into dilated natural fractures will cause a hump above the extrapolated straight line (line 3A-3B in Figure 3) on the GdP/dG derivative trace. The point where the GdP/dG data trace falls back on the straight line is the natural fracture opening pressure. The leakoff in the hump area is accelerated. Such leakoff behavior is classified pressure-dependent leakoff and can

cause problems with premature screenout during the main frac treatment.

3. Fracture height recession occurs when the fracture grows into a higher stress zone during pumping and then recedes during shut-in. The GdP/dG will sag below the straight line during the recession period. Fracture growth through zones of higher stress can cause a reduction in fracture width and early screenout.
4. Fracture-tip extension is evident when the GdP/dG lies on a straight line extrapolated above the origin. Fracture tip extension happens during the shut-in period. If this happens, it may be indicative that the formation is too tight to be productive.

In addition to identifying leakoff behavior, fracture width, and formation permeability, the GdP/dG derivative identifies fracture closure pressure at the point where the derivative drops down from a straight line extrapolated through the origin. Fracture closure pressure is used to calculate fluid efficiency input for the main frac treatment design. Bottom hole pressure minus closure pressure is used as a diagnostic plot of bottom hole pressure during the pumping of the main frac treatment. Additionally, the amount of closure stress will dictate the type of proppant to use.

Case 1: Westmoreland County, Pennsylvania

The Oriskany zone in this well was perforated from 8716' to 8734' with 2 shots per foot at 120° phasing for a total of 37, 0.39" holes. Gross interval was +/- 40 feet.

Figure 2 is a pressure (psi) and rate (bpm) vs. elapsed time (min) chart of the minifrac event for this well. Starting at elapsed time (etime) of 102 minutes, 20 bbl of fresh water and surfactant were pumped at +/- 2.5 bpm. Initial break was +/- 4900 psi with an average treating pressure of 4800 psi. The instantaneous shut-in pressure (ISIP) at the end of pumping was 4600 psi.

Figure 3 is a regression analysis using the GdP/dG derivative of the pressure falloff data from etime 110 to 122 minutes in figure 2. Point TC marks the closure time at 2.77 minutes at a surface pressure of 2600 psi. TC is determined to be at the point where the GdP/dG curve breaks over. Additionally, TC is at the intersection on lines 1B/1A and 2B/2A. These lines approximate the slope change of the Surface Pressure curve on the graph. Fluid efficiency can be calculated using the pressure change in GdP/dG from the ISIP to TC which in this instance is 0.24. Fluid efficiency should be +/- 0.5. This low fluid efficiency indicates a high fluid leak off into the formation. Case 1 is an example of pressure dependent leakoff (PDL) evidenced by the hump in the GdP/dG curve above the 3A-3B line. Fluid leakoff in this case is exacerbated by PDL. As a frac treatment is pumped, increased net pressure inside the fracture will increase the openings of the natural fractures. During pumping, the fluid leakoff will be higher than during shut-in pressure decline and the fluid efficiency can be overestimated. Overestimating fluid efficiency where $\%pad = 1 - \text{efficiency} / 1 + \text{efficiency}$ will result in smaller pads and early screenouts⁵. In this case where PDL has been identified, the calculated efficiency of

0.24 should be considered the maximum and the frac design should add some extra pad.

Figure 4 is a Horner plot from the pressure decline data. Extrapolation of line 1A-1B intersects the Y axis at 340 psi. Adding 340 psi to the hydrostatic head of 2990 psi equals 3330 psi for an estimated reservoir pressure. The known reservoir pressure for the Oriskany in this area is +/- 500 psi less than this. The Horner plot reservoir pressure estimation is only valid in pseudo-radial flow which does not occur until after fracture closure. The Horner plot is an indication that closure pressure may be found along the entire range of the GdP/dG derivative in Figure 3, in essence the plot gives an upper bound for looking for closure pressure.

Figure 5 is a net pressure plot of the subsequent frac treatment on this well. The net pressure is the bottom hole treating pressure (calculated from surface treating pressure) minus the closure pressure (6300 psi). The net pressure declines slightly until +/- 100 minutes, levels out, and then begins an increase at 140 minutes and the frac ends in a screenout. The net pressure for the entire job is abnormally high which contributes to the PDL problem.

Case 2: Westmoreland County, Pennsylvania

In this well both the Chert and the Oriskany were perforated and fraced. The Oriskany was perforated from 8876' to 8890' with 2 shots per foot for a total of 28 shots. The gross interval was 40 feet. The Chert was perforated in two places. The top perforation was at 8750' and the bottom perforation was at 8784'. There were a total of 48 shots in a gross interval of 110'.

Figure 6 is the rate/pressure chart for the breakdown and falloff of the Oriskany perforations. HCL acid (7-1/2%) was pumped at 2.9 bpm for +/- 4 minutes starting at etime of 12. The Oriskany appears to have some carbonate content as evidenced by the 1000 psi pressure drop while the pump rate was constant.

Figure 7 is a regression analysis using the GdP/dG derivative of the pressure falloff data from etime 17 to 37 minutes in figure 6. Point TC marks the closure time at 3.11 minutes at a surface pressure of 2775 psi. The hump of the GdP/dG derivative curve above the 3A-3B line indicates PDL is present and more severe than Case #1. Fluid efficiency is calculated to be 0.31. The per cent pad would be calculated as $1 - 0.31 / 1 + 0.31 = 52\%$ recommended for a frac treatment using water with no gel.

Figure 8 is the rate/pressure chart for the breakdown and falloff of the Chert perforations. HCL acid (7-1/2%) was pumped at 4.7 bpm for +/- 10 minutes starting at etime of 64. There is no evidence of any carbonate content in the Chert. The pressure response shows the initial breakdown of the rock followed by the slight decrease as the fracture is established and begins to grow.

Figure 9 is the regression analysis using the GdP/dG derivative of the pressure falloff data from etime 74 to 96 minutes in figure 8. Point TC marks the closure time at 5.08 minutes at a surface pressure of 2895 psi. The GdP/dG derivative hump above line 3A-3B indicates PDL here also. Fluid efficiency is calculated at 0.27 indicating an even greater fluid loss to the formation in the Chert.

Figure 10 is the net pressure plot (starting at an elapsed time of +/- 12 minutes) of the frac treatment pumped on this well. It was decided to treat the Oriskany and the Chert together with a 15 pounds per 1000 gallons (ppt) linear gel system at a minimum of 20 barrels per minute (bpm). 100 mesh sand was used as a fluid loss agent to help control the PDL into the natural fractures. Maximum 20/40 sand concentration was 1.5 pounds per gallon (ppg). Even with the use of 100 mesh sand there was not complete control of the fluid loss. As the net pressure increased, increased PDL exceeded the pump rate and a screen-out cut short the job in terms of total barrels to be pumped and pounds of sand designed to be placed into the zones. Radioactive tracer run with the sand on this job indicated a minimum created fracture height of 40 feet in the Oriskany and 45 feet in the Chert.

Case 3: Washington County, Ohio

This well was planned to be a 2 stage completion with the Oriskany being the second stage. The zone was perforated with 16 shots from 5163' to 5171'. Gross interval was +/-20 feet.

Figure 11 is the rate/pressure chart for the breakdown and falloff of the Chert perforations. 15% HCL acid was used to breakdown the perforations followed by pumping perforation sealer balls to insure breakdown of all perforations. Commencement of a step rate up followed by step rate down test began at 47 minutes etime lasting to 67 minutes etime. 130 barrels of treated water were pumped at rates from 2 bpm to 19 bpm and then back down to 2 bpm. The purpose of the step rate up test was to determine the rate at which a fracture was initiated. At the minimum rate of 2 bpm the fracture was already propagating. This is to be expected in a low permeability formation such as the Oriskany. Following the step rate back down, pumping stopped at 67 minutes etime and the pressure fall-off monitored for +/- 20 minutes.

Figure 12 is a Horner plot from the pressure decline data. Extrapolation of line 1A-1B intersects the Y axis at +/- 1200 psi. Addition to this 1200 psi of 2240 psi for the hydrostatic head gives an estimated reservoir pressure of 3440 psi. As with Case #1 the Horner plot indicates that closure pressure may be found along the entire range of the GdP/dG derivative.

Figure 13 is a regression analysis using the GdP/dG derivative of the 20 minutes pressure falloff data in figure 11. Point TC marks the closure time at 4.89 minutes at a surface pressure of 1726 psi. Leak off behavior here is quite different from Cases 1 and 2. There is no hump of the GdP/dG derivative above the 3A-3B line. There is no PDL. This is normal leakoff through the formation matrix.

Case 4: Washington County, Ohio

The Oriskany in this well was perforated from 5138' to 5142' with 16 shots. The gross interval was +/- 10 feet'. A treatment on the Oriskany of 2500 gal 15% HCL breakdown and ballout was pumped ahead of the step rate and pressure falloff seen in Figure 14.

Figure 15 is the regression analysis using the GdP/dG derivative of the pressure falloff data from etime 127 to 220 minutes in figure 14. Point TC marks the closure time at 52.6 minutes at a surface pressure of 678 psi. There appears to be

fracture-height recession. The GdP/dG derivative is sagging below line 3A-3B and the pressure curve is a slight concave down.⁶ The calculated fluid efficiency of 0.60 is much higher than seen in the first 3 cases. There is no leakoff into natural fractures in the interval and the matrix permeability is probably on the low side.

Conclusions

1. Cases 1 and 2 have similar pressure analyses which are different from the analyses of Cases 3 and 4. This is to be expected knowing that the geology of the Oriskany changes from east to west in the Appalachian Basin.
2. The importance of pre-frac diagnostics must be emphasized. The extra expenditure to perform the pump-in and falloff tests is a very small part of the cost of stimulation. The extra time to do the diagnostics can be minimized by doing the test the same day as the frac treatment.
3. The shut-in time to monitor pressure falloff is the most important part of the diagnostic procedure. The tight formations prevalent in the Appalachian Basin dictate long shut-in times to allow pressure bleed off to the point of fracture closure. A shut-in time of 1 hour or more is not unreasonable.
4. A contingency plan to deal with the effects of pressure dependent leakoff should be formulated if PDL is suspected. The best way to deal with PDL is to pump a large pad with 100 mesh sand.
5. The focus of this paper was use of GdP/dG derivative analysis of the pressure falloff after shut-in to determine closure pressure and leakoff behavior. The cases presented here show that picking TC is not an exact science and two analysts looking at the same data could present a strong argument for 2 different values of TC. The best application of GdP/dG derivative analysis will include along with it, step rate analysis, net pressure analysis, and history matching using a fracturing simulator.

Acknowledgements

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Figure 1. Extent of Oriskany Formation in the Appalachian Basin

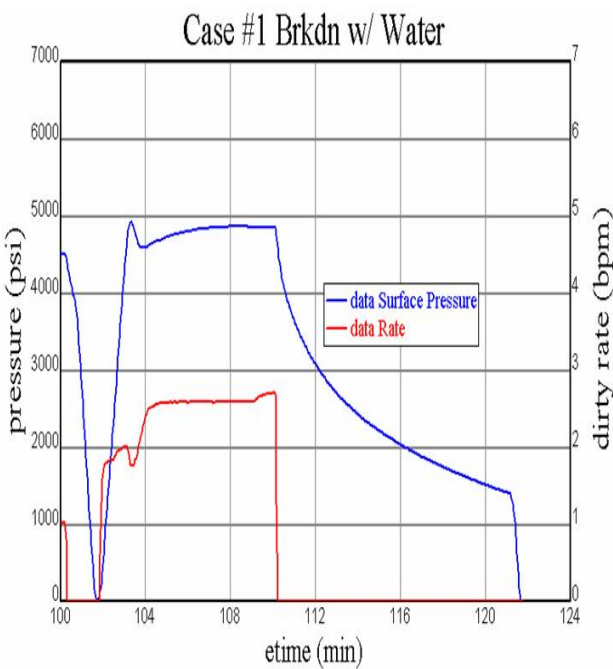


Figure 2. Perforation Breakdown and Pressure falloff

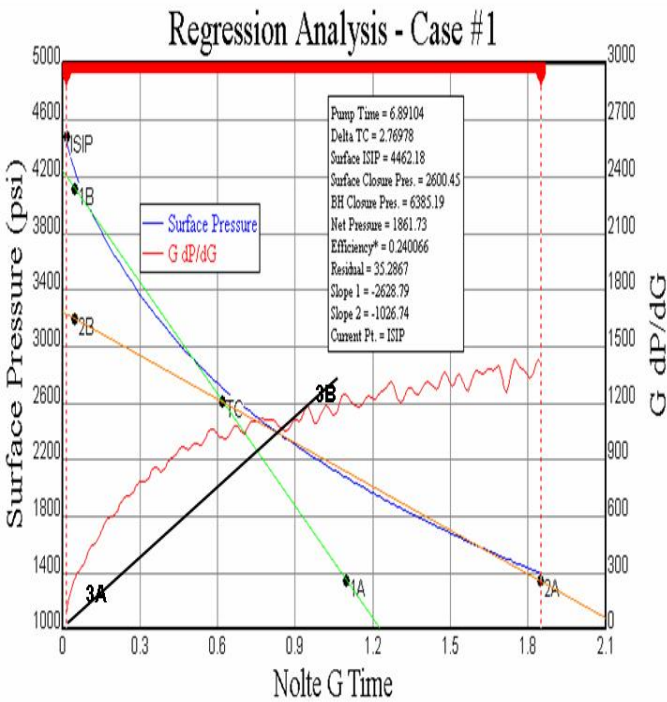


Figure 3. Pressure Falloff Analysis

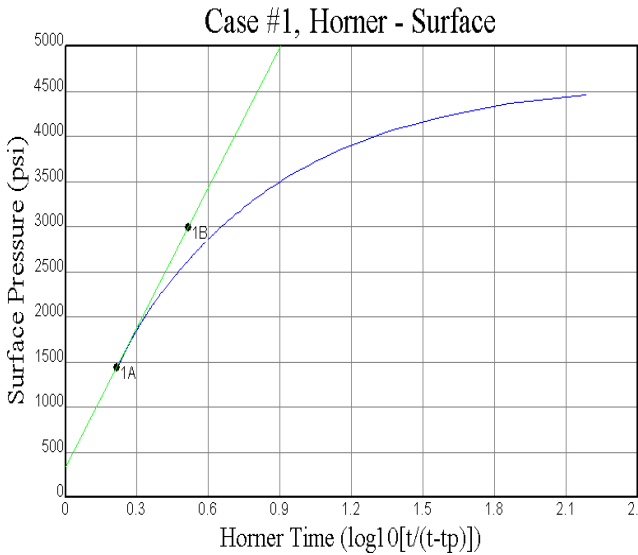


Figure 4. Horner Plot

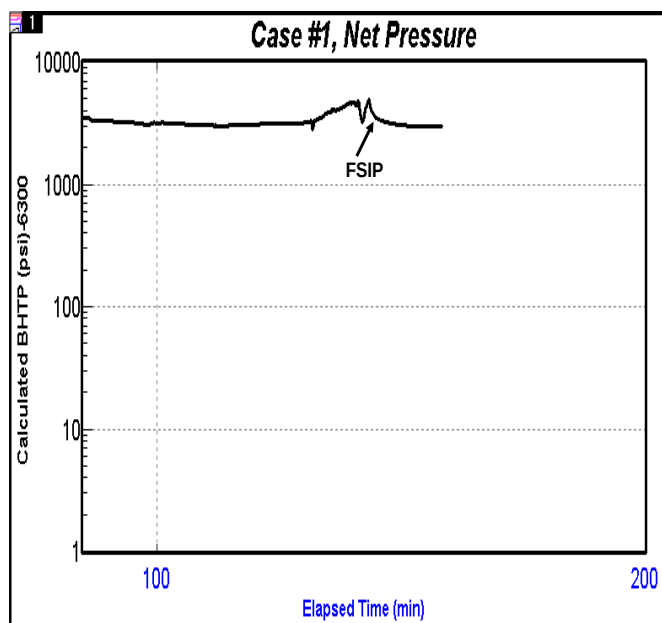


Figure 5. Net Pressure Plot of Frac Treatment

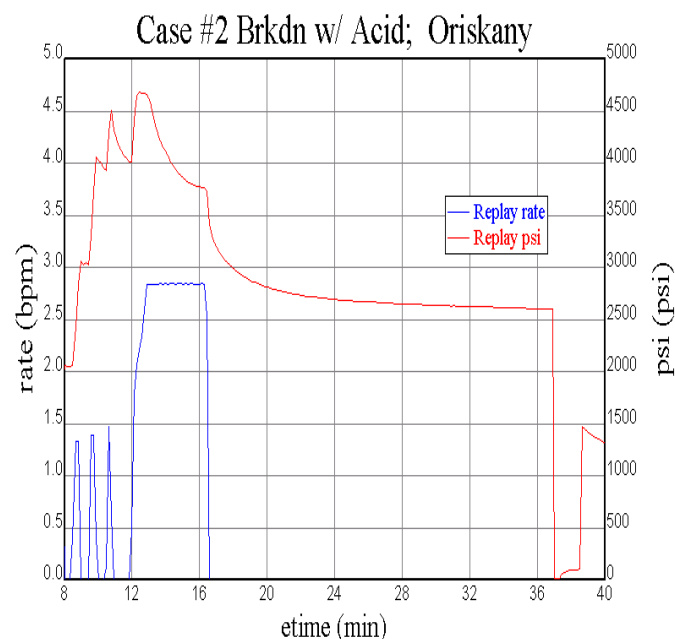


Figure 6. Oriskany Perforations Breakdown & Falloff

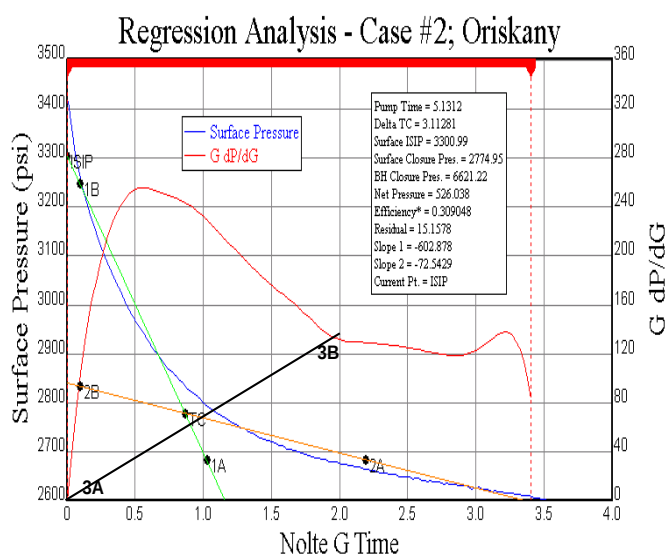


Figure 7. Pressure Falloff Analysis of Oriskany

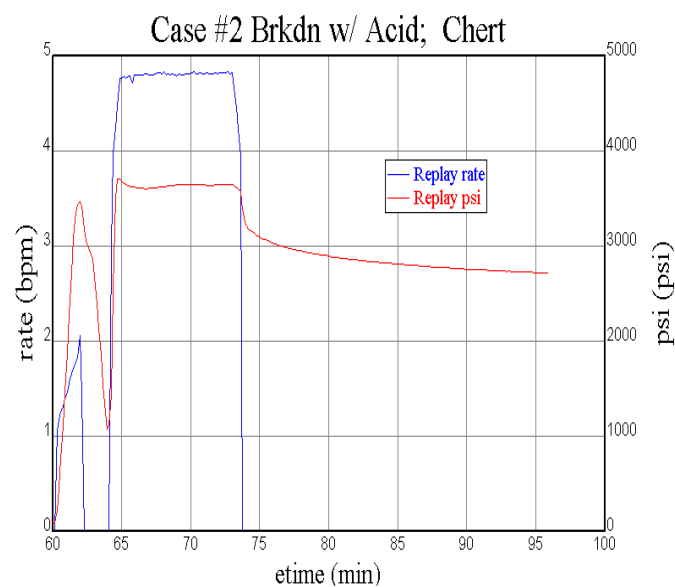


Figure 8. Chert Perforations Breakdown & Falloff

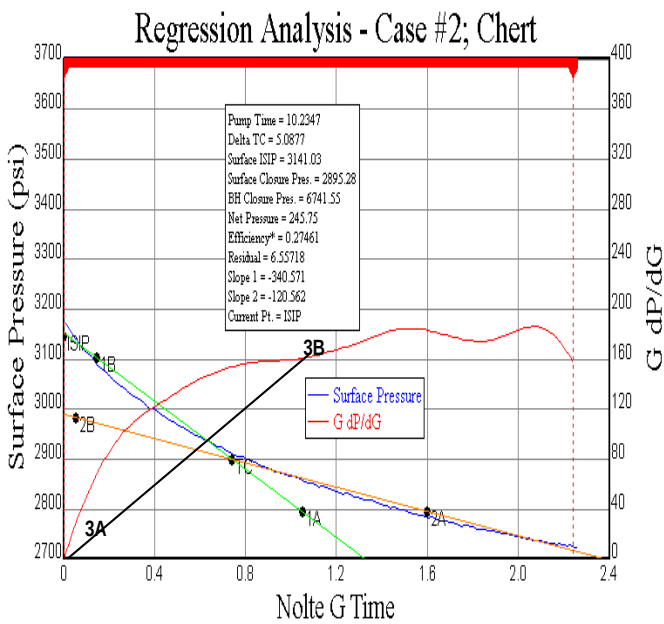


Figure 9. Pressure Falloff Analysis of Chert

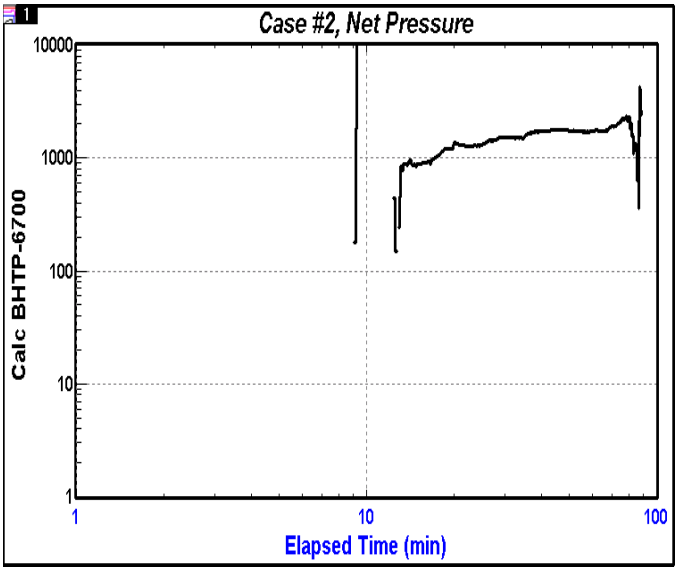


Figure 10. Net Pressure Plot of Frac Treatment on Chert/Oriskany

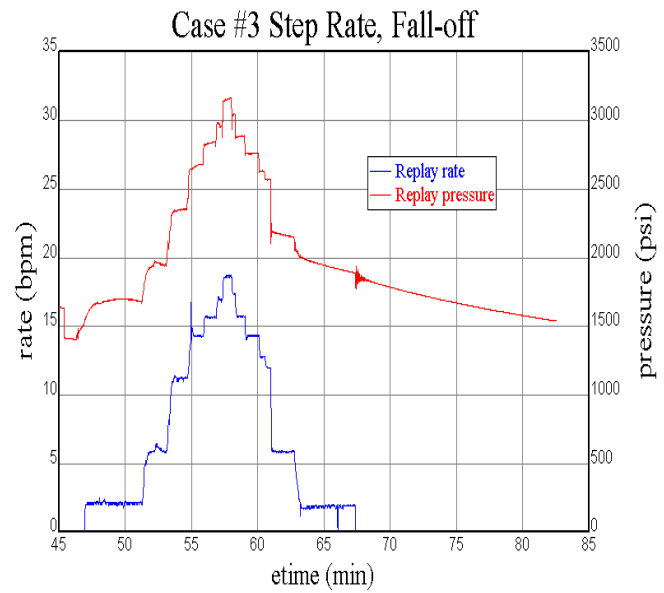


Figure 11. Step Rate & Falloff on the Oriskany

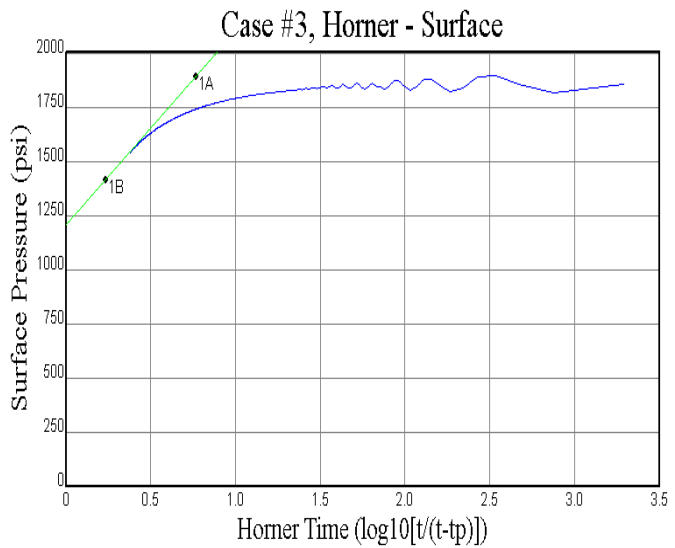


Figure 12. Horner Plot

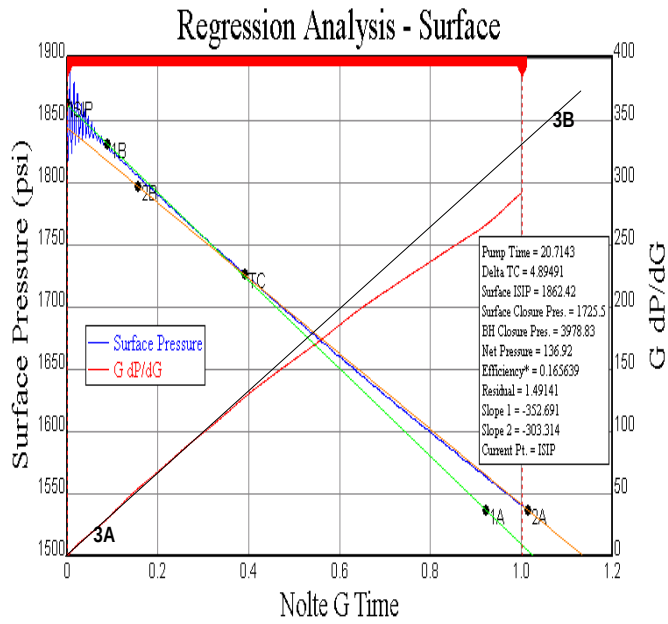


Figure 13. Pressure Falloff Analysis for Case #3

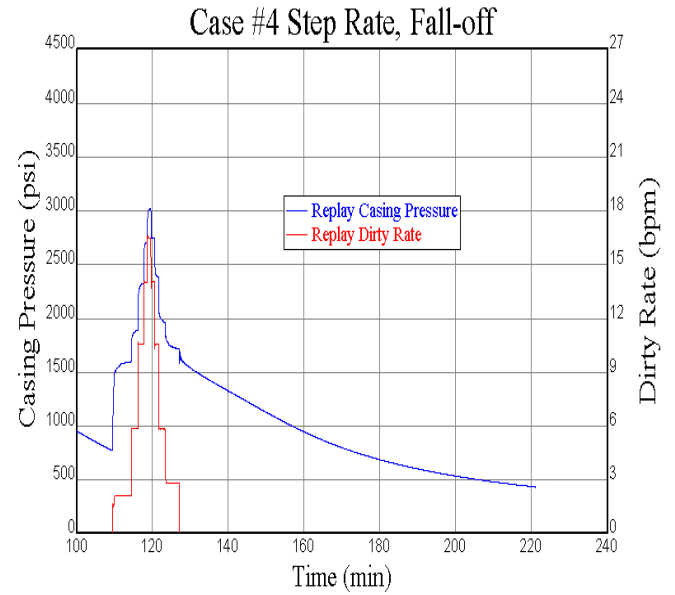


Figure 14. Step Rate and Falloff for Case #4

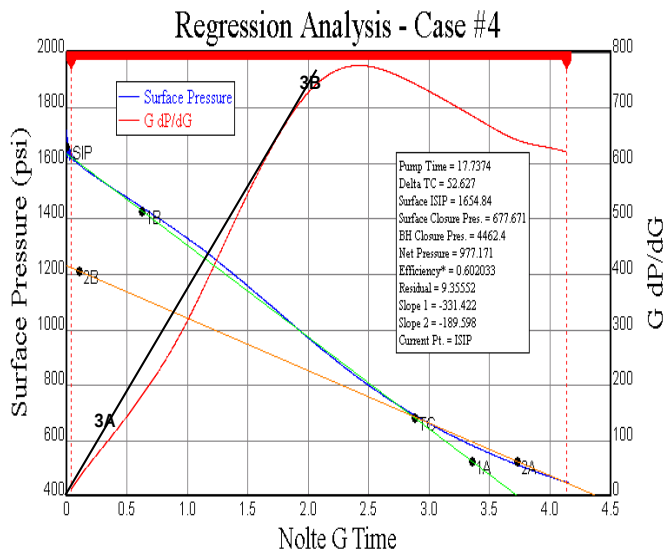


Figure 15. Pressure Falloff Analysis for Case #4